Algebra Monsoon 2026 Lecture 18 p-subgroups, Sylow's theorems

Algebra Monsoon 2026 Lecture 18

p-subgroups.

Let p be a prime number.

Def. A finite group P is said to be a *p-group* if $o(P)$ is a power of p.

Let G be a group. A subgroup $P \subset G$ is called a *p-subgroup* if P is a p-group.

Write $o(G) = p^n m$, with $m \not\equiv 0 \pmod{p}$.

A subgroup of order p^n is called a *p-Sylow*.

1) G has at least one p-Sylow.
2) Any two p-Sylows are conjugate in G
3) Every p-subgroup is contained a p-Sylow
4) If s is the number of p-Sylows, then
$$s \equiv 1 \pmod{p}$$
and $$m \equiv 0 \pmod{s}.$$

Clearly $s \equiv 1 \pmod{p} \Rightarrow s \geq 1$, so

1) is a special case of 4).

Let X be the set of subsets of G having p^n elements and s the number of subgroups of G of order p^n.

Lemma $o(X) \equiv s \pmod{p}$

Proof The group G acts on X by left translations

$$(g, A) \longmapsto \gamma_{g*}(A)$$

Let $S \subset X$ be a subset such that $S \to G\backslash X$ is a bijection, so that $X = \coprod_{A \in S} G.A$

Let G_A be the stabiliser of A, so that

$$o(G) = o(G.A).o(G_A).$$

Note that $o(G_A) \leq p^n$ for every $A \in X$.

Indeed, fixing an $a \in A$ gives an injection $g \mapsto g.a$ of G_A into A.

- If $o(G_A) < p^n$, then $o(G.A) \equiv 0 \pmod{p}$.
- If $o(G_A) = p^n$, then G_A is a p-Sylow.

Conversely, if $P \subset G$ is a p-Sylow, then $Pg \in X$ for every $g \in G$ and the stabiliser of Pg is P itself: it is clear that $P \subset G_{Pg}$; if $h \in G_{Pg}$, then $hg \in Pg$, so $h \in P$.

Similarly, if $P \subset G_A$ for some $A \in X$, then $PA \subset A$, and for every $a \in A$, we have $Pa \subset A$. Hence $A = Pa$, because $o(Pa) = o(A)$. Note the number of such A is $(G : P) = m$.

Now,
$$o(X) = \sum_{\substack{A \in S \\ o(G_A) < p^n}} o(G.A) + \sum_{\substack{A \in S \\ o(G_A) = p^n}} o(G.A).$$
$$\equiv 0 + sm \pmod{p},$$
hence the result. □

Corollary $s \equiv 1 \pmod{p}$, and hence $s > 0$.

Proof The lemma shows that $s \pmod{p}$ depends only on $o(G)$. Taking $G = \mathbb{Z}/p^n m\mathbb{Z}$, we get $s \equiv 1 \pmod{p}$.

Corollary If $o(G) \equiv 0 \pmod{p}$, then G has an element of order p.

Proof Let $P \subset G$ be a p-Sylow; we have $P \neq \{e\}$ because $o(G) \equiv 0 \pmod{p}$. If $x \in P$ is different from e, its order is p^m for some $m \geq 1$. Then $x^{p^{m-1}}$ has order p. □

Theorem 1) Every p-subgroup of G is contained in a p-Sylow.

2) Any two p-Sylows are conjugate in G.

Proof 1) Let $S \subset G$ be a p-Sylow and let $P \subset G$ be a p-subgroup. Take $X = G/S$ (the set of left translates of S), make G and P act upon it by left translations. We have $o(G/S) \not\equiv 0 \pmod{p}$, hence $o(X^P) \not\equiv 0 \pmod{p}$. In particular, there exists $A \in X$ such that $P.A = A$. Then $P \subset G_A$, and G_A is a conjugate of $G_S = S$ because A and S are in the same G-orbit (in fact $G.S = X$). This shows that P is contained in a p-Sylow (namely G_A).

Let S' be another p-Sylow. By 1),

2) Let $S' = gSg^{-1}$ for some $g \in G$. As they have the same order, $S' = gSg^{-1}$. □

It remains to prove that

$$m \equiv 0 \pmod{s}.$$

Let $N_G(S)$ be the normaliser of S in G. We have

$$(G : S) = (G : N_G(S))\,(N_G(S) : S)$$

$$= m \qquad = s$$

because the number of conjugates of a subgroup $H \subset G$ is equal to $(G : N_G(H))$.

Therefore $m \equiv 0 \pmod{s}$. □

Corollary $(G : N_G(S)) \equiv 1 \pmod{p}$.

Indeed, the number of G-conjugates of any subgroup $H \subset G$ is $(G : N_G(H))$, and we know that the number of conjugates of a p-Sylow is $\equiv 1 \pmod{p}$.

□